

Probabilistic Dataset Merge

Scenario 1: The Exact Match

What happens when the target row exists verbatim in the supplemental dataset

What this scenario demonstrates. This scenario asks a simple question: what does the system report when there is a perfect match — a supplemental row that is completely identical to the target in every feature? This is the best possible outcome and serves as our baseline. All signals should be at their ideal values: zero distance, zero ambiguity, and a confirmed symmetric match. If any signal were to fire under these conditions, something would be wrong with the system.

The Target Record

This is the record we are trying to match. The system will search the supplemental dataset for the row closest to these values.

Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units
2,469	40.2	99.5	649

Table 1: Target row values (raw, unstandardized).

All Candidate Matches

The table below shows all 20 supplemental rows, sorted from closest to farthest. Distances are computed in standardized space (see Section 3). The highlighted row is the best match — the one the system selects.

Rank	Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units	Distance
1	2,469	40.2	99.5	649	0.0000
2	2,948	33.4	98.5	687	1.0251
3	2,524	30.0	99.5	557	1.2511
4	3,885	37.2	100.0	1,004	1.4812
5	3,849	42.2	99.5	1,043	1.4969
6	4,158	29.8	99.4	1,089	2.1253
7	4,555	26.5	99.4	930	2.3420
8	2,512	25.8	96.1	521	2.4757
9	4,822	25.7	98.2	914	2.6008
10	5,094	27.3	100.0	1,041	2.6636
11	4,891	34.4	99.6	1,339	2.6822
12	5,006	30.1	97.5	1,230	2.8970
13	5,296	25.4	99.3	1,128	2.9913
14	4,223	36.3	94.2	1,132	3.3064
15	4,701	58.7	100.0	1,519	3.6746
16	6,303	29.8	100.0	1,490	3.8139
17	3,465	47.9	92.5	1,080	3.9509
18	6,516	23.3	99.3	1,385	4.0681
19	6,668	32.0	98.5	1,641	4.2209
20	7,389	35.0	100.0	1,851	4.8968

Note: “Distance” is the standardized Euclidean distance between the target and each candidate row. A distance of 0.0000 means a perfect, identical match.

How Distance Is Calculated: A Worked Example

We use the **rank 2** candidate below to illustrate the calculation step by step. (The rank 1 match has distance 0 and is trivial; rank 2 shows the full arithmetic.)

Step 1 — Put every feature on the same scale (standardization).

Raw values are measured in different units (counts, years, percentages). Before computing distance we convert each to a *standardized score*: how many standard deviations above or below the combined average that value sits.

$$z = \frac{\text{raw value} - \text{combined average}}{\text{combined spread}}$$

Feature	Target (raw)	Candidate (raw)	Combined Avg.	Combined Spread
Dragon Sightings	2,469	2,948	4463.95	1421.92
Avg. Wizard Age	40.2	33.4	33.87	8.34
Pct. Leprechaun Cottages	99.5	98.5	98.60	1.96
Goblin Family Units	649	687	1089.48	353.65

Applying the formula above to each feature gives these standardized scores:

Feature	Target (std.)	Candidate (std.)
Dragon Sightings	-1.4030	-1.0661
Avg. Wizard Age	+0.7595	-0.0560
Pct. Leprechaun Cottages	+0.4621	-0.0486
Goblin Family Units	-1.2455	-1.1381

Step 2 — Compute the squared difference for each feature, then sum and take the square root.

$$d = \sqrt{\sum_i (z_{\text{target},i} - z_{\text{cand},i})^2}$$

Feature	Target (std.)	Cand. (std.)	Squared difference
Dragon Sightings	-1.4030	-1.0661	0.1135
Avg. Wizard Age	+0.7595	-0.0560	0.6650
Pct. Leprechaun Cottages	+0.4621	-0.0486	0.2608
Goblin Family Units	-1.2455	-1.1381	0.0115
<i>Sum</i>			<i>1.0508</i>

$$d = \sqrt{1.0508} = 1.0251$$

Distance Distribution

The chart below shows the distance from the target to each of the 20 candidate rows, sorted closest to farthest. The **highlighted bar** is the selected match.

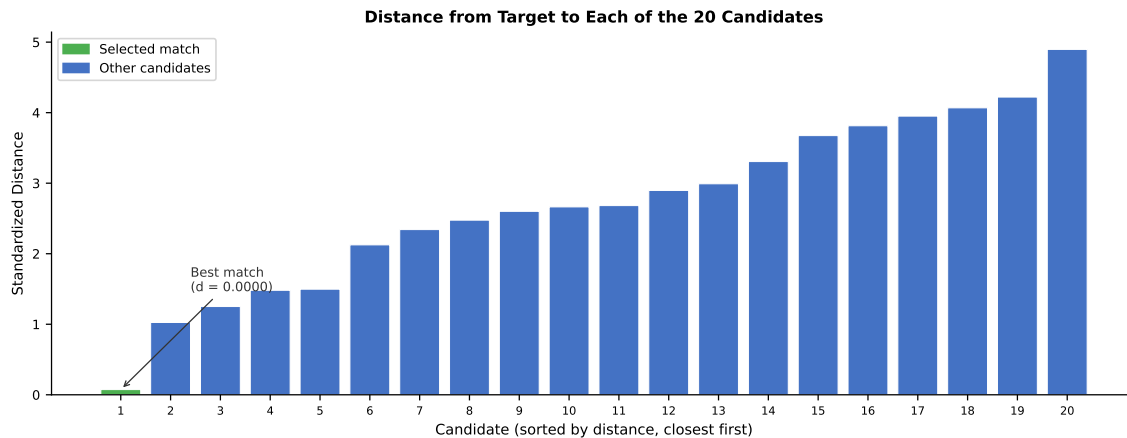


Figure 1: Distances from target to all 20 supplemental candidates, sorted ascending. The selected match is highlighted.

Signal Analysis

Each signal below is reported in the output file for every matched row. This section explains what each signal means and why it has the value it does for this particular match.

euc_distance 0.0000

The distance between the target and its best match is exactly **0.0000**. This means the two rows are identical in every feature — a perfect match.

per_feature_contribution (best match)

Because the distance is exactly 0, no feature contributed anything — all values are identical between the target and the matched row. The table shows n/a for all features.

Feature	Share of distance
Dragon Sightings	n/a (perfect match)
Avg. Wizard Age	n/a (perfect match)
Pct. Leprechaun Cottages	n/a (perfect match)
Goblin Family Units	n/a (perfect match)

nndr	0.0000
The Nearest Neighbor Distance Ratio (NNDR) compares the best match distance to the second-best match distance: d_1/d_2 . Because $d_1 = 0$, the ratio is $0 \div \text{anything} = 0$. A value of 0.0000 is the best possible score — no ambiguity whatsoever. No flag raised (threshold is 0.80).	
near_miss_count	0
The near-miss count checks how many supplemental rows have a ratio $d_1/d_i \geq 0.80$. Since $d_1 = 0$, every ratio is $0 \div d_i = 0$, well below the threshold. 0 near misses . No flag raised.	
mnn_confirmed	True
Mutual Nearest Neighbor (MNN) confirmation runs the search in reverse: does the matched supplemental row also point back to this target as its nearest neighbor? Since the matched row is an exact copy of the target, of course it does. Confirmed: True . This symmetry is a strong indicator of match quality.	
repeats	1
Only one supplemental row matched at distance 0 — the exact copy we inserted. If a second identical row had been present, repeats would be 2 and the tie flag would fire.	
smd (per feature)	[0.0000, 0.0000, 0.0000, 0.0000]
The Standardized Mean Difference (SMD) measures whether the distribution of the target dataset is systematically misaligned with its matches. With only one target row , there is no distribution to measure — SMD is undefined and reported as 0 for all features. In a real research run with many target rows, this signal would flag any feature that is consistently off across all matches.	
flags	(none)
No flags raised. This is the ideal outcome: perfect match, zero ambiguity, no ties, and a symmetric confirmation. Any well-functioning match on a row that exists verbatim in the supplemental dataset should produce this result.	

Data note: Values in this example are adapted from U.S. Census Bureau American Community Survey (ACS) public data. Column names have been replaced with illustrative labels for demonstration purposes. This document is intended for researcher education and internal system validation.